IT4010 Research Project – 2025 July

Topic Assessment Form

Project ID : 25-26J-438

1. Topic (12 words max)

AI-Powered History E-Learning System for Visually Impaired O-Level Students in Sri Lanka

2. Research group the project belongs to

CoEAI - Centre of Excellence for AI

3. Specialization of the project belongs to

Software Engineering (SE)

4. If a continuation of a previous project:

Project ID	-N/A-
Year	-N/A-

5. Brief description of the research problem including references (200 – 500 words max) – references not included in word count.

Visually impaired students in Sri Lanka face significant barriers in accessing quality education, particularly in subjects like History. While the Sri Lankan education system provides some support through Braille textbooks and audio recordings, these resources are often outdated, monotonous, and lack interactive or personalized engagement. In addition, most of the available material consists largely of text and is not sufficiently streamlined to grasp at the O-Level, especially for younger students who have extra learning difficulties.

The absence of adaptive and inclusive digital learning tools has widened the educational gap between sighted and visually impaired students. Current assistive technologies like screen readers or Braille books do not effectively support comprehension of complex historical narratives, nor do they offer interactivity, question-answer engagement, or summarization support that modern AI systems can enable. Furthermore, evaluating the academic work of visually impaired students - particularly in Braille - remains a manual, time-consuming process, prone to human error and inconsistencies in marking.

This research addresses the need for an AI-powered e-learning platform specifically designed to support O-Level History education for visually impaired students in Sri Lanka. By integrating text-to-speech (TTS) systems, natural language processing (NLP) for content simplification, voice-based interaction, smart summarization, and Braille sheet correction using AI, the proposed solution aims to offer a comprehensive and inclusive educational experience. The system will convert history textbooks into rich, interactive audio lessons, simplify content based on learning level, allow voice-based question answering, and automatically recognize and correct Braille answer sheets.

The study aims to explore the feasibility, effectiveness, and educational impact of applying state-of-the-art AI/ML technologies in this domain. The solution not only targets improving content accessibility and learning outcomes but also contributes toward achieving the UN's Sustainable Development Goal 4: Inclusive and equitable quality education for all. By focusing on history education - a subject that forms a foundational understanding of culture, identity, and civic responsibility - this research intends to empower visually impaired students with equal opportunities to succeed academically and socially.

6. Brief description of the nature of the solution including a conceptual diagram (250 words max)

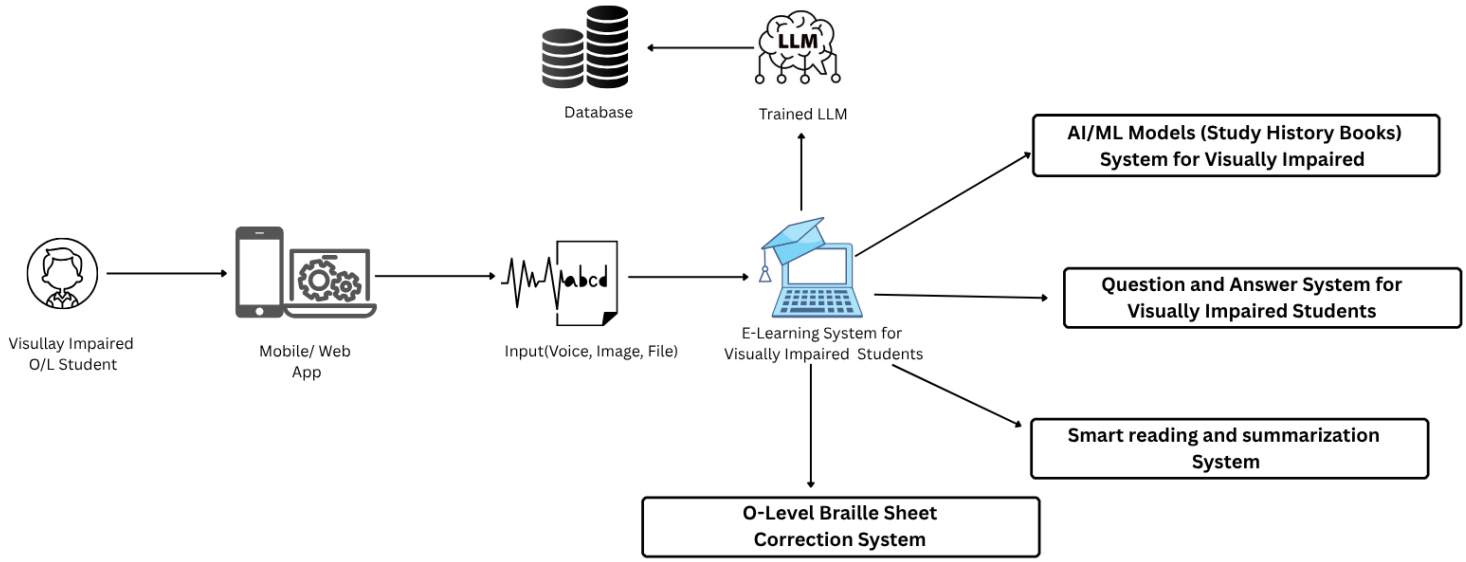
The proposed solution is an **AI-powered e-learning system** tailored for visually impaired O-Level students studying History. It is a multimodal platform that transforms traditional textbook learning into an **interactive audio-first learning experience** supported by assistive technologies.

The system consists of four integrated AI components:

1. **Audio-Based Learning Module** – Uses **Text-to-Speech (TTS)** with emotional tone to convert history textbooks into engaging audio lessons. **NLP techniques** simplify historical texts into age-appropriate narratives.
2. **Smart Summarization Engine** – Analyzes external reading materials such as **magazines, newspapers, and books**, and provides **extractive or abstractive summaries** based on reading levels and auditory needs.
3. **Voice-Based Question and Answer System** – Allows students to ask syllabus-related questions via **Speech-to-Text (STT)**. Answers are generated using **AI-driven knowledge models** and delivered via voice.
4. **Braille Sheet Correction System** – Accepts scanned Braille answer sheets and converts Braille dots into readable text using **Braille OCR**. It supports English Braille standards.

The system is accessible through mobile and desktop devices with **screen reader compatibility**, audio navigation. It empowers visually impaired students by making history learning more engaging, accessible, and autonomous.

Conceptual Diagram



7. Brief description of specialized domain expertise, knowledge, and data requirements (300 words max)

This research requires an interdisciplinary combination of expertise in artificial intelligence, accessibility technologies, education, and Sri Lankan history. The project's core depends on developing intelligent systems that can accurately process and deliver history content in formats accessible to visually impaired students.

Specialized Domain Expertise Required:

- **Artificial Intelligence & Machine Learning**: Expertise in Natural Language Processing (NLP), Text-to-Speech (TTS), Speech-to-Text (STT), and transformer-based models is essential for building the learning, summarization, and question-answering modules.
- **Computer Vision**: Knowledge in Braille dot detection, image preprocessing, and Optical Character Recognition (OCR) is required for the Braille sheet correction system.
- **Accessibility & Inclusive Design**: Familiarity with assistive technologies (e.g., screen readers, Braille standards, audio navigation) is necessary to ensure the platform aligns with the needs of visually impaired students.
- **Software Development and UX/UI Design**: Skilled software developers and UX/UI designers are needed to build a user-friendly, engaging, and efficient e-learning platform.
- **Educational Pedagogy**: Understanding how visually impaired learners engage with historical narratives is key to simplifying and structuring content meaningfully.

Knowledge Requirements:

- O-Level **Sri Lankan History curriculum**.
- **Braille standards** for English languages.
- Existing **assistive learning tools** and accessibility compliance guidelines.

Data Requirements:

- Digital versions of **O-Level History textbooks**.
- Sample materials from **magazines, newspapers, and books**.
- Recorded or scanned **Braille answer sheets** from past O-Level exams for training and testing.
- Voice interaction datasets for STT and TTS personalization.

Collaboration with educators, institutions for the blind, and education boards is essential to access curriculum materials and validate the platform's usability and impact.

8. Objectives and Novelty

Main Objective The main objective of this research is to develop and implement a comprehensive AI-powered e-learning system specifically designed for visually impaired O-Level students studying History in Sri Lanka. This research aims to transform traditional textbook-based learning into an interactive, accessible, and engaging audio-first educational experience through the integration of advanced assistive technologies. The main objective encompasses not only technological innovation but also social impact through educational equity promotion, making it a comprehensive research initiative that bridges technology, education, and social responsibility.			
Member Name with Registration No	Sub Objective	Tasks	Novelty
IT22232090 Sathushan T	AI-powered system that reads and summarizes content from magazines, newspapers, and books,, voice-based summaries tailored for visually impaired students	Identify and categorize resource types: magazines, newspapers, and books. Develop a user-guided system to ask follow-up actions based on the selected resource. Implement extractive and abstractive summarization models tailored to student reading level and content complexity. Convert text to expressive speech using TTS for visually impaired learners Provide options to translate summaries into local languages(Sinhala and Tamil).	This system uniquely combines **adaptive summarization**, **interactive decision-making**, and **multimodal accessibility** tailored specifically for **visually impaired students**. It allows students to control the summarization process and delivers personalized, reading-level-sensitive outputs for effective learning. Unlike existing tools, this solution **recognizes and differentiates between resource types** (magazines, newspapers, and books) and adapts its summarization strategy accordingly. This ensures the most

			effective presentation and delivery of content based on the structure, tone, and complexity of each resource type. Unlike traditional summarization tools, this solution integrates **interactive decision-making** where users guide how the summarization process unfolds.
IT22238344 Abishaan T	Develop an AI-powered system to accurately recognize and convert scanned English Braille O-Level exam sheets into digital text for automated evaluation.	Collect scanned English Braille sheets and annotate dot positions with corresponding text labels for model training. Apply denoising, skew correction, and dot alignment to improve image quality and accuracy for dot detection. Use computer vision models to detect and locate Braille dots in the images. Convert detected dot patterns into English text using a Braille OCR engine. Apply AI language models to resolve ambiguities and ensure accurate	Focuses exclusively on O-Level English Braille exam sheets, addressing real-world challenges like noisy, skewed, handwritten, or printed documents. Combines advanced computer vision and OCR techniques with AI-powered natural language processing for context-sensitive Grade 2 contraction expansion. First known system to automate Braille answer sheet correction specifically for O-Level exams, reducing examiner workload and improving accessibility.

		translation and expansion of Braille content.	Tailors language model integration to historical and exam-specific vocabulary, enhancing accuracy in subject-specific content transcription.
IT22219084 Senavirathna B.M.P.C	Develop an AI-driven multi-sensory storytelling module with haptic feedback to deliver Sri Lankan history lessons in an engaging and accessible manner for visually impaired students.	Design and train an NLP-based AI model to generate vivid, descriptive audio narratives of historical events (e.g., Anuradhapura era, Sigiriya construction) using Sri Lankan history texts. Create culturally relevant audio scripts with sound effects (e.g., temple bells, battle sounds) to enhance immersion. Integrate haptic feedback devices (e.g., wearable gloves) to synchronize tactile sensations (e.g., vibrations for battle scenes) with audio narratives. Evaluate the module's effectiveness through user testing with visually impaired students to measure engagement and comprehension.	The integration of multi-sensory storytelling with haptic feedback is a novel approach in educational systems for visually impaired students. By combining descriptive audio narratives with synchronized tactile feedback, the module creates an immersive learning experience that allows students to "feel" historical events, enhancing emotional engagement, spatial understanding, and long-term retention of Sri Lankan history concepts. This approach is unique as it leverages affordable haptic devices and culturally contextualized content, addressing accessibility and engagement gaps in traditional history education.
IT22604330 Dolawattage K.J	Build a Voice-based question and answer system with history curriculum	Develop Sri Lankan O-level history curriculum knowledge base.	This component focuses on the creation of an intelligent question-answering (QA) system specifically

	integration for visually impaired O/Level students.	Implement a variety of MCQs based on O-Level history syllabus. Create voice-optimized interaction patterns. Build contextual follow-up question generation system. Design culturally relevant historical knowledge retrieval.	trained on Sri Lankan O-level history curriculum. Designing with voice-optimized interaction at its core, providing an inclusive and accessible learning experience for students with visual impairments. By incorporating culturally relevant content and aligning with the national syllabus, the system ensures educational relevance while addressing a critical accessibility gap in mainstream learning platforms.

9. Individual component description of how it is complied with the specialization.

Member Name with Registration No	Description
IT22232090 Sathushan T	This component applies software engineering principles in building an intelligent, modular system with NLP, TTS, and user-driven interaction logic. It demonstrates system design, API integration, and accessibility-focused UI/UX, addressing complex user requirements.
IT22238344 Abishaan T	Involves the application of software engineering in image processing, OCR integration, and AI model development. It includes dataset handling.
IT22219084 Senavirathna B.M.P.C	This component applies software engineering to develop an AI-driven multi-sensory storytelling module with haptic feedback for visually impaired O-Level students learning Sri Lankan history. Using NLP, it converts history texts into audio narratives synchronized with haptic feedback via wearable devices. Iterative development and user testing ensure reliability and engagement, supporting inclusive education.
IT22604330 Dolawattage K.J	Illustrates software architecture design for voice interfaces and intelligent agents. Emphasizes structured data modeling, knowledge base construction, and voice interaction patterns. It ensures software maintainability and curriculum-focused knowledge delivery through structured system workflows.

10. Supervisor details

	Title	First Name	Last Name	Signature
Supervisor	Mr	Suresh	Jemanda	2025/6/26
Co-Supervisor	Ms.	Kaushalya	Rajapakse	KR 2025/06/26
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

This part is to be filled by the Topic Screening Staff members.

a) Does the chosen research topic possess a comprehensive scope suitable for a final-year project?

Yes		No	

b) Does the proposed topic exhibit novelty?

Yes		No	

c) Do you believe they have the capability to successfully execute the proposed project?

Yes		No	

d) Do the proposed sub-objectives reflect the students' areas of specialization?

Yes		No	

e) Supervisor's Evaluation and Recommendation for the Research topic:

IT4010 Research Project – 2025 July Topic Assessment Form

Acceptable: Mark/Select as necessary

Topic Assessment Accepted	
Topic Assessment Accepted with minor changes*	
Topic Assessment to be Resubmitted with major changes*	
Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

Staff Member's Name	Signature

*__Important__:

1. According to the comments given by the evaluator, make the necessary modifications and get the approval by the **Evaluator**.
2. If the project topic is rejected, identify a new topic, and request the RP Team for a new topic assessment.